

NLP-Based Business Insights from Customer Feedback (Python)

Project Overview

This project focuses on extracting actionable business insights from unstructured customer feedback using Natural Language Processing (NLP). Instead of limiting analysis to basic sentiment scores, the project identifies recurring customer pain points, evaluates their business impact, and translates findings into clear recommendations for decision-makers.

The goal is to demonstrate how a data analyst can convert raw text data into structured insights that support product, operations, and customer experience teams.

Business Problem

Organizations receive large volumes of unstructured customer feedback through reviews, surveys, and support channels. Manually analyzing this data is not scalable, and simple sentiment analysis often fails to explain **why** customers are dissatisfied.

★ **This project answers:**

- What are customers actually talking about?
 - Which issues generate the most negative sentiment?
 - Where should the business prioritize improvement efforts?
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Tools & Technologies

- ★ Pandas
- ★ NumPy
- ★ NLTK
- ★ Scikit-learn
- ★ Matplotlib

★ Seaborn

Dataset

A realistic synthetic customer feedback dataset was created to simulate:

- ★ Product reviews
- ★ App feedback
- ★ Support-related complaints

Each record includes:

- ★ Customer feedback text
- ★ Rating (1–5)
- ★ Channel/source of feedback

The dataset intentionally contains unstructured, noisy text similar to real-world data.

Analysis Workflow

1 Text Cleaning & Preprocessing

- ❖ Lowercasing text
- ❖ Removing punctuation and stopwords
- ❖ Preparing clean text for analysis

2 Exploratory Text Analysis

- ❖ Identified most frequent complaint terms
- ❖ Compared language used in positive vs negative feedback
- ❖ Analyzed feedback volume across channels

3 Topic Modeling (LDA)

- ❖ Applied Latent Dirichlet Allocation (LDA) to uncover key themes
- ❖ Identified dominant topics such as:
 - Payment & refund issues
 - App performance problems
 - Order & delivery concerns
 - Customer support experience

4 Sentiment & Impact Analysis

- ❖ Mapped customer ratings to sentiment categories
- ❖ Analyzed sentiment distribution by topic
- ❖ Created a priority score combining issue volume and severity

5 Business Recommendations

- ❖ Ranked issue themes by business impact
 - ❖ Developed targeted action plans for product, operations, and support teams
 - ❖ Suggested KPIs to monitor improvement over time
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Key Insights

- ★ Customer complaints are highly concentrated around a small number of themes.
 - ★ Payment/refund failures and app performance issues generate the strongest negative sentiment and should be treated as top priorities.
 - ★ Order-related issues are frequent but less severe, indicating opportunities for communication improvements rather than systemic fixes.
 - ★ Positive feedback tends to be generic and offers limited actionable guidance.
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Final Conclusion & Business Recommendations

- ★ By combining exploratory text analysis, topic modeling, and sentiment-based prioritization, this project demonstrates how unstructured customer feedback can be transformed into data-driven business decisions.
 - ★ Focusing improvement efforts on the highest-impact themes—particularly payment stability and app performance—can significantly enhance customer satisfaction with targeted effort. This project showcases an end-to-end NLP analytics approach designed for real-world business decision-making rather than purely technical experimentation.
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Future Enhancements

- ★ Automate topic monitoring over time
- ★ Integrate sentiment trend alerts
- ★ Build a dashboard for leadership review

★ Apply the approach to real customer review datasets

Execution Environment

The entire project was developed and executed using **Google Colab**, ensuring cloud-based accessibility, reproducibility, and ease of experimentation.

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